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Longitudinal changes in strength, power, speed, and body mass in American football players: Diminishing returns as training age increases

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Abstract

The aims of this study were to: 1) detail the strength, power, speed, and body mass (i.e. physical qualities) of National Collegiate Athletic Association (NCAA) Division I American football players by playing position across a four-year collegiate career, and 2) quantify the rate of change in physical qualities as athletes progress through their eligibility period. 2628 observations from 512 NCAA Division I American football players were collected during a standardised testing battery that took place across a 15-year period. One-repetition maximum bench press and back squat, 40-yard sprint, vertical jump, standing broad jump, and body mass were analyzed. Year-on-year changes in physical characteristics by the entire cohort and positional groups were analyzed using linear mixed models with Cohen's effect size (ES) ± 95% confidence intervals (CI). Improvements in physical qualities occurred over the four years, although the largest changes were evident in the first year, with the improvements becoming less pronounced each year. For example, across the cohort, *large* effects in the bench press (ES ±95%CI: 1.63 ± 0.18) and back squat (ES ±95%CI: 1.62 ± 0.18) occurred in the first year but only *small* changes occurred between years 3 and 4 (ES ±95%CI: 0.45 ± 0.16 and 0.26 ± 0.16, respectively). Collectively, this study demonstrates the longitudinal changes in strength, power, speed, and body mass of collegiate Division I American football players across different positional groups. Furthermore, it shows that physical qualities improve throughout the four years, but the year-on-year change becomes smaller each year and therefore suggests physical qualities become harder to increase/improve.

Keywords

Athlete development, back squat, bench press, body mass, jump, National Collegiate Athletic Association, sprint

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Introduction

American football is an intermittent team sport, where athletes are required to sprint, rapidly change direction, tackle, and have physical contact with opposition players.[1] The sport has several unique positions that have distinct physical characteristics and experience different physical demands.[1] For example, wide receivers and running backs are often lighter and faster than offensive and defensive linemen, who are required to block or tackle opposition players.[2,3] Alternatively, other positions (e.g. tight ends) often require a mix of physical size, strength, and speed. With the disparate demands between positional groups,[3] there is a need to identify and develop specific physical characteristics for athletes within these different playing positions. For instance, if a pre-season testing battery identifies a lineman with low absolute strength but fast sprint times, future training could be tailored to enhance the respective physical qualities relevant for the position. Furthermore, it is also important to understand how these qualities change over time as they have been shown to be important discriminators between professional and non-professional players and athletes that start matches at the collegiate level compared to those that do not.[4,5]

In the United States of America, talented American football players can be recruited to play for a college for 1–5 years. At the collegiate level, players are exposed to programmes that provide dedicated strength and conditioning and nutritional support. These formalised training environments often allow for substantial improvements in physical capacity.[6,7] While previous research by Jacobson et al.,[6] has demonstrated changes in collegiate American football player physical qualities, broad groupings of different playing positions (i.e. 'linemen' and 'skill players') were used. However, by including a range of different positions in each group, this mitigates our understanding of changes in physical characteristics of each position. Hoffman et al.,[8] examined anthropometric and physical performance variables of National Collegiate Athletic Association (NCAA) Division III college football players, although whether the findings are applicable to Division I players, is unclear due to the distinct physical qualities typically observed between by higher and lower level NCAA athletes. Additionally, while Secora et al.,[9] provided an overview of physical qualities and performance data pertaining to a range of different playing positions, the acute study design (i.e. physical qualities of athletes in 1987 compared to 2000) does not provide understanding of how players change and the expected magnitudes that can occur by year. The rate with which physical change occurs often decreases as time passes (e.g. athletes may achieve large improvements in upper body strength throughout their freshman year but only achieve smaller improvements during their senior year)[6,10] which is commonly referred to as the "law of diminishing returns", but this phenomenon has not been investigated by playing position in American football athletes.

American football is a high-intensity collision sport that requires an array of well-developed physical qualities. These qualities can be targeted through structured strength and conditioning interventions. However, there is currently a lack of normative data detailing the physical qualities of these athletes by position and how they change over time. Thus, the aims of this study were to: 1) detail the strength, power, speed, and body mass (i.e. physical qualities) of NCAA Division 1 American football players by playing position across a four-year collegiate career, and 2) quantify the rate of change in physical qualities as athletes progress through their eligibility period.

Methods

Research design

To detail and quantify the changes in strength, power, speed, and body mass that occur across a 4-year collegiate playing career, a retrospective analysis of 15 years of performance data from athletes in a Division 1 collegiate football programme was completed. Athletes were grouped into 11 positional categories (i.e. corner, defensive ends, defensive tackle, linebacker, offensive linemen, quarterbacks, receiver, running backs, safety, specialists, and tight ends) and undertook standardised testing at the start of each year. Players were allocated to a designated positional group at the beginning of the study and remained within that group for the duration. The bench press and back squat were used to assess maximal strength. The broad jump and vertical jump were used as tests of lower body power. Forty-yard sprint was used to assess sprint performance. Additionally, body mass was recorded.

Subjects

Across a 15-year time frame, data were collected from 512 NCAA Division 1 male collegiate football athletes, aged 18–23 years old, who participated in 1 university's athletic performance training programme. Across the testing timepoints, 2628 testing observations were made. The athletes were all evaluated in the off-season prior to the new season. No athletes under the age of 18 years were included in the study, and the protocol was approved by the University of Missouri Institutional Review Board in accordance with the Declaration of Helsinki. Written informed consent was obtained from all athletes.

Procedures

All tests were administered by the university's athletic performance training staff, who were Certified Strength and Conditioning Specialists. The procedures for the testing

battery were consistent for each year and were conducted in a sequential manner described below. For each test, athletes were required to wear a team-issued T-shirt and pair of shorts. When assessing body mass, players did not wear shoes.

Bench press and back squat. The bench press was performed in a standardised manner consistent with National Strength and Conditioning Association (NSCA) guidelines with five points of contact.[11] The barbell was required to be lowered to the chest but not be bounced off the chest and returned to the starting position by straightening the elbows. A standardised loading protocol was used, beginning with warm-up sets of 40–80% of estimated 1RM, progressing to single attempts with corresponding load increments of ~5–10%. Rest intervals of 2–5 min were provided between attempts, with load adjusted as needed until the 1RM or heaviest load lifted was achieved. The bench press one repetition maximum (1RM) was based upon 5 or fewer repetitions to technical failure. If 1 repetition was collected, that was set as the 1RM. If 2–5 repetitions were completed, then the Epley equation was used to predict 1RM, which has shown a standard error of estimate between 6.38 kg and 7.08 kg.[12] All attempts were monitored by a qualified strength and conditioning staff member.

The back squat was evaluated by a strength and conditioning staff member to ensure that depth was of an acceptable level, with the crease of the hip below the top of the knee, and a neutral spinal position was maintained.[11] A standardised protocol similar to the process outlined above for the bench press was used, except load increments were approximately 10–20% of estimated 1RM. If depth or technique were not maintained during a repetition, this would result in either a cessation of the test or the repetition not counting towards the 1RM prediction. Similar to the bench press, up to a five repetition maximum was completed. If 1 repetition was collected, that was set as the 1RM, and if 2–5 repetitions were completed, the Epley equation was used to estimate 1RM.[12] Both tests have a coefficient of variation <5%.[13]

Broad and vertical jump. Similar to previous research using the standing broad jump,[14] the assessment started with the athletes standing with their toes behind a line marked on the floor. From this line, a measuring tape was situated along both sides of the lane. From a standing position, athletes were instructed to use a countermovement technique by flexing their knees to a self-selected depth and an accompanying coordinated arm motion to jump as far as possible and to land on two feet (i.e. stick the landing). The distance between the rear of the start line and the heel during the landing was measured. If an athlete landed in a staggered stance, the distance was measured to the rearmost heel. If an athlete failed to stick the landing, the result was discarded, and an additional attempt was provided until two successful attempts had been completed, with the greatest distance recorded. To ensure accuracy, a coach would mark the landing point, the athlete would step away, and the coach would use a metre stick to ensure that both sides of the tape said the same number. Each trial was separated by approximately 1 min, for recovery.

The vertical jump was performed using a Just Jump mat (Just Jump System, Probotics, Huntsville, Alabama, USA). Initially, the athletes stood still on the mat. When instructed, they jumped as high as possible into the air, using arm swing, and landing with legs that were at near full extension. This technique was used to ensure that numbers were not artificially inflated, as flight time underpins the Just Jump mat's estimation of jump height (i.e. tucking the legs would extend flight time and thus inflate the estimated jump height). If the strength and conditioning member that assessed the jumps deemed that 'tucking' or landing in a deep squat occurred, the repetition was not recorded, and the athlete was instructed to repeat the effort. Two attempts were provided for each athlete, with the greatest height from the device recorded. Both jump tests have reported coefficients of variation <5%.[15]

40-yard sprint. All sprint assessments were performed on an indoor artificial turf surface (Indoor Field Turf; FieldTurf, Montreal, CA), with the athletes instructed to dress in standard team-issued T-shirt, shorts, and cleats. The athletes completed 2 trials separated by a recovery period of at least 5 min. The athletes were instructed to assume a 3-point stance for the starting position (i.e. 1 hand and 2 feet in contact with the ground).[16] An electronic timing device (SpeedTrap, Model II; Brower Timing Systems, Draper, UT) was used to measure sprint times, with the athlete instructed to place one hand on the device's touchpad.[17] Once the athlete initiated the start of their sprint, they moved their hand from the touchpad, which triggered the electronic timing device to begin timing the sprint. The timing ended once the athlete broke an infra-red beam placed 75 cm above the ground at the 40-yard mark. The best effort from the 2 attempts was used.

Body mass. Each athlete's body mass was recorded using a standard physician's beam scale (Mettler Toledo, Columbus, OH). Within-day reliability of the scales was found to be within 1%, which aligns with accepted standards for repeated measures in athletic populations. The athletes stood on the scale and were instructed to remain motionless until the beam indicator was aligned and stable with the centre mark. In accordance with the university's athletic performance training programme's policy, body mass was recorded to the nearest 1 lb and then converted to kilograms.

Statistical analysis

All statistical analyses were conducted in RStudio (R Core Team, Vienna, Austria; version 4.4.2) using the

lme4, emmeans, effectsize, and *dplyr* packages. Linear mixed models were employed to examine changes in physical characteristics across an athlete's NCAA career. Data were first pre-processed to ensure consistency, including unit conversions (e.g. body mass and strength measures from pounds to kilograms; vertical jump height from inches to centimetres) and appropriate variable coding.

To examine the year-over-year changes in physical characteristics by positional group, a linear mixed model was specified with year and position as a fixed effect and Athlete ID as a random intercept to account for repeated measures within individuals. Estimated marginal means were extracted using the *emmeans* package, and pairwise comparisons between years within each position were adjusted using Tukey's post hoc correction. Second, overall changes (i.e. for the entire cohort) across years were completed to assess longitudinal changes across the entire cohort. Again, linear mixed models were fitted, but these models excluded position as a fixed effect. The *emmeans* package was again used to extract outcomes for each year, and year-over-year differences were analyzed using Tukey-adjusted pairwise comparisons.

Effect sizes were computed from t-values using the *effectsize* package, and confidence intervals for Cohen's d were computed for each contrast. Effect sizes were interpreted using thresholds of 0.2, 0.6, 1.2, and 2.0 to signify small, moderate, large, and very large effects, respectively.[18] Statistical significance was set at $p < 0.05$.

Results

Descriptive data and between year contrasts for each positional group and the entire cohort combined can be found in Tables 1–6. Body mass by position and across years are presented in Table 1. Bench press and back squat results are provided in Tables 2 and 3, respectively. The vertical jump, broad jump, and 40-yard dash outcomes are available in Tables 4, 5, and 6. To support the visualisation of these data, Figure 1 provides positional trends and changes for the entire cohort for the bench press, back squat, vertical jump, and 40-yard sprint. Effect size changes for the entire cohort can be found in Figure 2. Of note, the effect size change (e.g. change in physical quality from year 1 to year 2) was greater in the first year compared to the following years for each outcome measure. Specifically, large changes in strength measures and moderate changes in body mass, 40-yard sprint, and jump performances were observed. Subsequent years all showed substantially reduced rates of improvement (Figure 2). Furthermore, a similar trend was observed for each positional group with the first year providing the greatest effect size change.

Discussion

The aims of this study were to: 1) detail the strength, power, speed, and body mass (i.e. physical qualities) of NCAA Division 1 American football players by playing position across a four-year collegiate career, and 2) quantify the rate of change in physical qualities as athletes progress through their eligibility period. While there are distinct differences in the physical qualities of athletes by playing position, there are noticeable decreases in the year-on-year improvements in strength, power, speed, and body mass of American football players as they progress throughout a collegiate programme. For instance, in the back squat, the effect size decreases from $d = 1.62 \pm 0.18$ in year 1 to year 2 ($p < 0.001$) to $d = 0.45 \pm 0.16$ in year 2 to year 3 ($p < 0.001$). Indeed, the magnitude of change in the first year is often substantially greater than the changes that occur in subsequent years. This provides evidence that as an athlete's training age increases, diminishing returns occur. Thus, coaches need to temper their expectations of how much improvement will likely occur. This is important for helping to set expectations for both athletes and coaches alike.

There are substantial changes in physical performance as athletes progress through a Division 1 NCAA football programme. Furthermore, athletes involved in the line of scrimmage tend to be substantially stronger and have greater body mass than all other positions. However, these positions also often tend to be the slowest in the 40-yard sprint and achieve lower values in the lower-body power tests. Alternatively, corners and receivers tend to have the lowest body mass but have the greatest jump performances and sprint times. These findings support previous research that has investigated differences in sprint momentum and change of direction performance.[4] Nonetheless, it is clear that different positional groups have differing physical requirements, with this study providing the most comprehensive overview of physical qualities in American football players by positional group at the NCAA level. This information can be used by coaches and aspiring athletes as reference data to help guide athlete development and support training prescription.

There is a noticeable decrease in the rate of development as athletes progress throughout their eligibility period at college. Figure 2 highlights these changes with the bench press and back squat demonstrating large effect sizes from year 1 to year 2, with small to moderate changes in subsequent years. These trends are prevalent in all other physical qualities, with the greatest increases in body mass, sprint, and jump performance occurring in the first year. These marked differences in the rate of change across years can likely be attributed to the law of diminishing returns.[10,19] Previous research[10] has demonstrated the decline in improvement in youth rugby league athletes as a consequence of training age (i.e. the number of years that an individual has been exposed to structured strength

Table 1. Body mass of NCAA Division I American football players over a 4-year collegiate career by position.

Position	Year 1 Est. ± 95% CL (n)	Year 2 Est. ± 95% CL (n)	Year 3 Est. ± 95% CL (n)	Year 4 Est. ± 95% CL (n)	Between year contrasts Est. ± 95% CL; *p* value; d ± 95% CL
Body mass (kg)					
Corner	82.73 ± 2.33 (43)	83.87 ± 2.43 (32)	84.3 ± 2.64 (18)	85.51 ± 2.9 (11)	Y1 - Y2 = 1.14 ± 1.8; p = 0.36; d = 0.12 ± −0.15
					Y2 - Y3 = 0.43 ± 2.27; p = 0.96; d = 0.04 ± −0.15
					Y3 - Y4 = 1.21 ± 2.88; p = 0.7; d = 0.08 ± 0.15
Defensive Ends	107.89 ± 2.45 (39)	110.53 ± 2.57 (28)	111.78 ± 2.82 (15)	109.64 ± 3.12 (9)	Y1 - Y2 = 2.64 ± 1.92; p ≤ 0.001; d = 0.26 ± 0.15
					Y2 - Y3 = 1.25 ± 2.48; p = 0.56; d = 0.10 ± 0.15
					Y3 - Y4 = −2.15 ± 3.18; p = 0.3; d = −0.13 ± 0.15
Defensive Tackles	122.84 ± 2.48 (38)	127.27 ± 2.52 (34)	129.08 ± 2.64 (23)	130.2 ± 2.84 (14)	Y1 - Y2 = 4.43 ± 1.75; p ≤ 0.001; d = 0.49 ± 0.15
					Y2 - Y3 = 1.81 ± 2.04; p = 0.1; d = 0.17 ± 0.15
					Y3 - Y4 = 1.12 ± 2.55; p = 0.67; d = 0.08 ± 0.15
Linebacker	99.25 ± 1.96 (61)	101.45 ± 2.09 (39)	102.09 ± 2.22 (26)	104.9 ± 2.43 (16)	Y1 - Y2 = 2.2 ± 1.72; p ≤ 0.001; d = 0.26 ± 0.14
					Y2 - Y3 = 0.63 ± 2.04; p = 0.92; d = 0.06 ± 0.15
					Y3 - Y4 = 2.81 ± 2.54; p = 0.02; d = 0.23 ± 0.15
Offensive Line	129.13 ± 1.61 (90)	132.36 ± 1.67 (69)	134.7 ± 1.75 (49)	136.97 ± 2 (23)	Y1 - Y2 = 3.23 ± 1.3; p ≤ 0.001; d = 0.50 ± 0.15
					Y2 - Y3 = 2.33 ± 1.49; p ≤ 0.001; d = 0.32 ± 0.15
					Y3 - Y4 = 2.27 ± 2.06; p = 0.02; d = 0.23 ± 0.15
Quarterback	94.97 ± 3.12 (24)	96.66 ± 3.25 (18)	96 ± 3.48 (11)	97.79 ± 4.06 (5)	Y1 - Y2 = 1.69 ± 2.4; p = 0.27; d = 0.13 ± 0.15
					Y2 - Y3 = −0.67 ± 2.93; p = 0.94; d = −0.04 ± 0.15
					Y3 - Y4 = 1.79 ± 4.14; p = 0.68; d = 0.08 ± 0.15
Receiver	86.34 ± 1.87 (67)	87.86 ± 1.97 (46)	88.64 ± 2.09 (31)	89.59 ± 2.3 (18)	Y1 - Y2 = 1.52 ± 1.59; p = 0.07; d = 0.19 ± 0.14
					Y2 - Y3 = 0.78 ± 1.87; p = 0.78; d = 0.09 ± 0.15
					Y3 - Y4 = 0.95 ± 2.38; p = 0.81; d = 0.08 ± 0.15
Running Back	90.53 ± 2.55 (36)	90.75 ± 2.66 (26)	92.75 ± 2.83 (17)	92.17 ± 3.38 (7)	Y1 - Y2 = 0.22 ± 1.98; p = 0.99; d = 0.02 ± 0.15
					Y2 - Y3 = 2.01 ± 2.4; p = 0.14; d = 0.16 ± 0.15
					Y3 - Y4 = −0.58 ± 3.47; p = 0.97; d = −0.03 ± 0.15
Safety	88.6 ± 2.19 (49)	90.52 ± 2.32 (32)	91.34 ± 2.46 (22)	92.21 ± 2.75 (12)	Y1 - Y2 = 1.92 ± 1.79; p = 0.03; d = 0.20 ± 0.14
					Y2 - Y3 = 0.82 ± 2.09; p = 0.74; d = 0.08 ± 0.15
					Y3 - Y4 = 0.87 ± 2.72; p = 0.84; d = 0.06 ± 0.15
Specialist	87.32 ± 2.89 (28)	90.16 ± 3.03 (20)	91.49 ± 3.22 (13)	92.71 ± 3.43 (9)	Y1 - Y2 = 2.84 ± 2.27; p = 0.01; d = 0.24 ± 0.15
					Y2 - Y3 = 1.33 ± 2.71; p = 0.58; d = 0.09 ± 0.15
					Y3 - Y4 = 1.21 ± 3.24; p = 0.77; d = 0.07 ± 0.15
Tight End	104.4 ± 2.75 (31)	106.35 ± 2.9 (21)	108.68 ± 3.05 (15)	111.56 ± 3.65 (6)	Y1 - Y2 = 1.95 ± 2.22; p = 0.11; d = 0.17 ± 0.14
					Y2 - Y3 = 2.33 ± 2.55; p = 0.09; d = 0.18 ± 0.15
					Y3 - Y4 = 2.88 ± 3.74; p = 0.19; d = 0.15 ± 0.15
All	101.44 ± 1.64 (511)	103.73 ± 1.65 (370)	105.07 ± 1.67 (241)	106.3 ± 1.72 (130)	Y1 - Y2 = 2.29 ± 0.21; p ≤ 0.001; d = 0.78 ± 0.15
					Y2 - Y3 = 1.34 ± 0.25; p ≤ 0.001; d = 0.39 ± 0.15
					Y3 - Y4 = 1.23 ± 0.33; p ≤ 0.001; d = 0.27 ± 0.14

Est: Estimated marginal means; 95% CL: 95% confidence limit; *n*: sample; *d*: Cohen's *d* effect size; Y1: year 1; Y2: year 2; Y3: year 3; Y4: year 4.

Table 2. Bench press of NCAA Division I American football players over a 4-year collegiate career by position.

Position	Year 1 Est. ± 95% CL (n)	Year 2 Est. ± 95% CL (n)	Year 3 Est. ± 95% CL (n)	Year 4 Est. ± 95% CL (n)	Between year contrasts Est. ± 95% CL; *p* value; d ± 95% CL
Bench Press (kg)					
Corner	128.14 ± 5.23 (40)	137.45 ± 5.45 (30)	141.32 ± 5.98 (17)	146.65 ± 6.53 (11)	Y1 - Y2 = 9.31 ± 4.3; p ≤ 0.001; d = 0.46 ± 0.16
					Y2 - Y3 = 3.87 ± 5.43; p = 0.26; d = 0.15 ± 0.16
					Y3 - Y4 = 5.33 ± 6.66; p = 0.17; d = 0.17 ± 0.16
Defensive Ends	149.48 ± 5.45 (36)	157.68 ± 5.78 (25)	166.32 ± 6.59 (12)	158.28 ± 8.27 (5)	Y1 - Y2 = 8.2 ± 4.82; p ≤ 0.001; d = 0.36 ± 0.16
					Y2 - Y3 = 8.64 ± 6.31; p ≤ 0.001; d = 0.29 ± 0.16
					Y3 - Y4 = −8.04 ± 9.63; p = 0.14; d = −0.18 ± 0.16
Defensive Tackles	158.74 ± 5.45 (36)	171.16 ± 5.55 (32)	175.5 ± 6.09 (17)	181.48 ± 6.79 (10)	Y1 - Y2 = 12.43 ± 4.25; p ≤ 0.001; d = 0.62 ± 0.17
					Y2 - Y3 = 4.33 ± 5.33; p = 0.16; d = 0.17 ± 0.16
					Y3 - Y4 = 5.99 ± 7.12; p = 0.13; d = 0.18 ± 0.16
Linebacker	144.2 ± 4.32 (58)	156.72 ± 4.62 (38)	160.68 ± 5.01 (24)	160.91 ± 6.6 (8)	Y1 - Y2 = 12.52 ± 4.04; p ≤ 0.001; d = 0.69 ± 0.16
					Y2 - Y3 = 3.96 ± 4.9; p = 0.18; d = 0.18 ± 0.16
					Y3 - Y4 = 0.23 ± 7.83; p = 1; d = 0.01 ± 0.16
Offensive Line	151.94 ± 3.61 (80)	162.71 ± 3.79 (59)	171.08 ± 4.04 (40)	181.51 ± 4.88 (17)	Y1 - Y2 = 10.78 ± 3.31; p ≤ 0.001; d = 0.73 ± 0.17
					Y2 - Y3 = 8.37 ± 3.87; p ≤ 0.001; d = 0.49 ± 0.16
					Y3 - Y4 = 10.42 ± 5.61; p ≤ 0.001; d = 0.42 ± 0.16
Quarterback	124.39 ± 7.33 (20)	133.93 ± 7.65 (15)	138.93 ± 8.76 (7)	145.4 ± 10.8 (3)	Y1 - Y2 = 9.55 ± 6.16; p ≤ 0.001; d = 0.33 ± 0.16
					Y2 - Y3 = 5 ± 8.49; p = 0.43; d = 0.12 ± 0.16
					Y3 - Y4 = 6.47 ± 12.22; p = 0.52; d = 0.11 ± 0.16
Receiver	123.77 ± 4.11 (66)	132.71 ± 4.44 (40)	137.92 ± 4.9 (23)	144.5 ± 5.48 (14)	Y1 - Y2 = 8.94 ± 3.64; p ≤ 0.001; d = 0.52 ± 0.16
					Y2 - Y3 = 5.21 ± 4.67; p = 0.02; d = 0.24 ± 0.16
					Y3 - Y4 = 6.58 ± 5.97; p = 0.02; d = 0.24 ± 0.16
Running Back	144.85 ± 5.56 (36)	152.68 ± 6.13 (19)	159.52 ± 6.5 (14)	162.71 ± 7.91 (6)	Y1 - Y2 = 7.83 ± 5.19; p ≤ 0.001; d = 0.32 ± 0.16
					Y2 - Y3 = 6.84 ± 6.66; p = 0.04; d = 0.22 ± 0.16
					Y3 - Y4 = 3.19 ± 8.88; p = 0.79; d = 0.08 ± 0.16
Safety	129.74 ± 4.78 (48)	139.44 ± 5.18 (29)	141.93 ± 5.57 (19)	144.22 ± 7.49 (6)	Y1 - Y2 = 9.7 ± 4.33; p ≤ 0.001; d = 0.47 ± 0.16
					Y2 - Y3 = 2.49 ± 5.24; p = 0.61; d = 0.10 ± 0.16
					Y3 - Y4 = 2.29 ± 8.89; p = 0.91; d = 0.05 ± 0.16
Specialist	111.62 ± 6.42 (27)	126.14 ± 6.77 (19)	128.9 ± 7.53 (10)	136.73 ± 8.42 (6)	Y1 - Y2 = 14.52 ± 5.34; p ≤ 0.001; d = 0.58 ± 0.16
					Y2 - Y3 = 2.76 ± 6.93; p = 0.73; d = 0.09 ± 0.16
					Y3 - Y4 = 7.83 ± 9.13; p = 0.12; d = 0.18 ± 0.16
Tight End	139.11 ± 5.99 (31)	149.61 ± 6.45 (19)	154.5 ± 6.89 (13)	159.01 ± 8.14 (6)	Y1 - Y2 = 10.5 ± 5.27; p ≤ 0.001; d = 0.42 ± 0.16
					Y2 - Y3 = 4.89 ± 6.45; p = 0.21; d = 0.16 ± 0.16
					Y3 - Y4 = 4.51 ± 8.71; p = 0.54; d = 0.11 ± 0.16
All	138.5 ± 1.91 (483)	148.99 ± 1.99 (330)	154.37 ± 2.11 (197)	159.4 ± 2.4 (92)	Y1 - Y2 = 2.29 ± 0.21; p ≤ 0.001; d = 1.63 ± 0.18
					Y2 - Y3 = 1.34 ± 0.25; p ≤ 0.001; d = 0.68 ± 0.16
					Y3 - Y4 = 1.23 ± 0.33; p ≤ 0.001; d = 0.45 ± 0.16

Est: Estimated marginal means; 95% CL: 95% confidence limit; *n*: sample; *d*: Cohen's *d* effect size; Y1: year 1; Y2: year 2; Y3: year 3; Y4: year 4.

Table 3. Back squat of NCAA Division I American football players over a 4-year collegiate career by position.

Position	Year 1 Est. ± 95% CL (n)	Year 2 Est. ± 95% CL (n)	Year 3 Est. ± 95% CL (n)	Year 4 Est. ± 95% CL (n)	Between year contrasts Est. ± 95% CL; *p* value; d ± 95% CL
Back Squat (kg)					
Corner	190.54 ± 9.54 (43)	210.26 ± 10.38 (30)	217.12 ± 13.93 (10)	214.64 ± 13.93 (10)	Y1 - Y2 = 19.72 ± 10.2; p ≤ 0.001; d = 0.41 ± 0.16 Y2 - Y3 = 6.87 ± 16.03; p = 0.69; d = 0.09 ± 0.16 Y3 - Y4 = −2.49 ± 19.52; p = 0.99; d = −0.03 ± 0.16
Defensive Ends	212.37 ± 10.32 (35)	240.8 ± 11.11 (26)	246.51 ± 12.97 (14)	234.02 ± 18.15 (5)	Y1 - Y2 = 28.43 ± 11.43; p ≤ 0.001; d = 0.52 ± 0.16 Y2 - Y3 = 5.71 ± 14.24; p = 0.73; d = 0.09 ± 0.16 Y3 - Y4 = −12.5 ± 22.48; p = 0.48; d = −0.12 ± 0.17
Defensive Tackles	229.16 ± 10.26 (36)	265.44 ± 10.77 (29)	268.32 ± 12.04 (18)	282.27 ± 14.59 (9)	Y1 - Y2 = 36.27 ± 10.5; p ≤ 0.001; d = 0.74 ± 0.17 Y2 - Y3 = 2.88 ± 13; p = 0.94; d = 0.05 ± 0.16 Y3 - Y4 = 13.95 ± 17.56; p = 0.17; d = 0.17 ± 0.17
Linebacker	211.07 ± 8.27 (56)	237.99 ± 9.42 (33)	243.44 ± 10.88 (19)	254.38 ± 13.37 (10)	Y1 - Y2 = 26.92 ± 10.4; p ≤ 0.001; d = 0.57 ± 0.16 Y2 - Y3 = 5.44 ± 13.09; p = 0.79; d = 0.10 ± 0.16 Y3 - Y4 = 10.94 ± 18.24; p = 0.47; d = 0.14 ± 0.16
Offensive Line	215.47 ± 6.8 (81)	245.18 ± 7.35 (58)	258.2 ± 8.11 (39)	271.8 ± 11.56 (13)	Y1 - Y2 = 29.71 ± 7.93; p ≤ 0.001; d = 0.84 ± 0.17 Y2 - Y3 = 13.02 ± 9.45; p ≤ 0.001; d = 0.32 ± 0.17 Y3 - Y4 = 13.6 ± 15.15; p = 0.1; d = 0.20 ± 0.16
Quarterback	174.8 ± 13.34 (22)	196.46 ± 14.86 (14)	213.9 ± 17.8 (7)	225.78 ± 23.44 (3)	Y1 - Y2 = 21.67 ± 14.87; p ≤ 0.001; d = 0.31 ± 0.16 Y2 - Y3 = 17.43 ± 19.92; p = 0.11; d = 0.19 ± 0.16 Y3 - Y4 = 11.89 ± 29.38; p = 0.72; d = 0.09 ± 0.17
Receiver	181.28 ± 7.76 (65)	200.38 ± 8.92 (36)	208.01 ± 10.15 (22)	211.89 ± 11.69 (14)	Y1 - Y2 = 19.09 ± 9.12; p ≤ 0.001; d = 0.44 ± 0.16 Y2 - Y3 = 7.64 ± 11.67; p = 0.33; d = 0.14 ± 0.16 Y3 - Y4 = 3.87 ± 14.89; p = 0.91; d = 0.06 ± 0.16
Running Back	213.94 ± 10.56 (34)	233.6 ± 11.55 (23)	238.13 ± 12.82 (15)	255.19 ± 16.13 (7)	Y1 - Y2 = 19.66 ± 11.88; p ≤ 0.001; d = 0.35 ± 0.16 Y2 - Y3 = 4.53 ± 14.5; p = 0.85; d = 0.07 ± 0.16 Y3 - Y4 = 17.06 ± 20.13; p = 0.13; d = 0.18 ± 0.16
Safety	190.2 ± 9.06 (47)	215.15 ± 10.38 (27)	221.94 ± 11.35 (19)	226.29 ± 14.74 (8)	Y1 - Y2 = 24.94 ± 10.71; p ≤ 0.001; d = 0.49 ± 0.16 Y2 - Y3 = 6.79 ± 12.73; p = 0.52; d = 0.12 ± 0.16 Y3 - Y4 = 4.35 ± 18.26; p = 0.93; d = 0.05 ± 0.16
Specialist	156.72 ± 11.9 (27)	178.49 ± 12.79 (20)	187.82 ± 14.46 (12)	205.14 ± 16.76 (7)	Y1 - Y2 = 21.77 ± 12.71; p ≤ 0.001; d = 0.36 ± 0.16 Y2 - Y3 = 9.33 ± 15.48; p = 0.41; d = 0.13 ± 0.17 Y3 - Y4 = 17.32 ± 19.91; p = 0.11; d = 0.19 ± 0.17
Tight End	188.69 ± 11.62 (29)	215.01 ± 13.13 (17)	236.26 ± 14.68 (11)	243.28 ± 17.71 (6)	Y1 - Y2 = 26.32 ± 13.27; p ≤ 0.001; d = 0.42 ± 0.16 Y2 - Y3 = 21.25 ± 16.82; p = 0.01; d = 0.27 ± 0.16 Y3 - Y4 = 7.01 ± 22.01; p = 0.84; d = 0.07 ± 0.16
All	199.3 ± 3.37 (479)	224.87 ± 3.64 (318)	233.71 ± 4.05 (187)	240.63 ± 4.89 (92)	Y1 - Y2 = 25.57 ± 3.47; p ≤ 0.001; d = 1.62 ± 0.18 Y2 - Y3 = 8.84 ± 4.36; p ≤ 0.001; d = 0.45 ± 0.16 Y3 - Y4 = 6.92 ± 6.08; p = 0.02; d = 0.26 ± 0.16

Est: Estimated marginal means; 95% CL: 95% confidence limit; *n*: sample; *d*: Cohen's *d* effect size; Y1: year 1; Y2: year 2; Y3: year 3; Y4: year 4.

Table 4. Vertical jump of NCAA Division I American football players over a 4-year collegiate career by position.

Position	Year 1 Est. ± 95% CL (n)	Year 2 Est. ± 95% CL (n)	Year 3 Est. ± 95% CL (n)	Year 4 Est. ± 95% CL (n)	Between year contrasts Est. ± 95% CL; *p* value; d ± 95% CL
Vertical Jump (cm)					
Corner	90.43 ± 2.59 (42)	94.09 ± 2.76 (30)	90.5 ± 3.38 (12)	92.68 ± 3.46 (11)	Y1 - Y2 = 3.66 ± 2.48; p ≤ 0.001; d = 0.31 ± 0.16 Y2 - Y3 = −3.59 ± 3.57; p = 0.05; d = −0.21 ± 0.16 Y3 - Y4 = 2.18 ± 4.37; p = 0.57; d = 0.11 ± 0.16
Defensive Ends	80.13 ± 2.76 (35)	83.56 ± 2.96 (25)	84.42 ± 3.35 (14)	82.5 ± 4.26 (6)	Y1 - Y2 = 3.42 ± 2.85; p = 0.01; d = 0.24 ± 0.16 Y2 - Y3 = 0.86 ± 3.49; p = 0.92; d = 0.05 ± 0.16 Y3 - Y4 = −1.92 ± 5.11; p = 0.77; d = −0.08 ± 0.16
Defensive Tackles	72.16 ± 2.76 (36)	74.39 ± 2.84 (31)	73.47 ± 3.15 (18)	74.31 ± 3.62 (10)	Y1 - Y2 = 2.23 ± 2.48; p = 0.1; d = 0.19 ± 0.16 Y2 - Y3 = −0.92 ± 3.03; p = 0.86; d = −0.06 ± 0.16 Y3 - Y4 = 0.84 ± 4.11; p = 0.95; d = 0.04 ± 0.16
Linebacker	84.44 ± 2.24 (53)	86.76 ± 2.43 (36)	88.7 ± 2.7 (22)	87.84 ± 3.12 (13)	Y1 - Y2 = 2.32 ± 2.5; p = 0.08; d = 0.20 ± 0.16 Y2 - Y3 = 1.94 ± 2.95; p = 0.37; d = 0.15 ± 0.16 Y3 - Y4 = −0.86 ± 3.91; p = 0.97; d = −0.05 ± 0.16
Offensive Line	67.41 ± 1.83 (82)	69.62 ± 1.94 (60)	70.04 ± 2.09 (41)	71.9 ± 2.73 (15)	Y1 - Y2 = 2.21 ± 1.89; p = 0.01; d = 0.26 ± 0.16 Y2 - Y3 = 0.42 ± 2.2; p = 0.99; d = 0.04 ± 0.16 Y3 - Y4 = 1.86 ± 3.38; p = 0.56; d = 0.12 ± 0.16
Quarterback	83.06 ± 3.55 (22)	85.48 ± 3.77 (16)	88.94 ± 4.27 (9)	87.06 ± 5.29 (4)	Y1 - Y2 = 2.42 ± 3.37; p = 0.25; d = 0.15 ± 0.16 Y2 - Y3 = 3.46 ± 4.42; p = 0.18; d = 0.16 ± 0.16 Y3 - Y4 = −1.88 ± 6.38; p = 0.87; d = −0.06 ± 0.16
Receiver	88.57 ± 2.07 (66)	92.26 ± 2.26 (42)	93.55 ± 2.63 (21)	91.02 ± 2.74 (18)	Y1 - Y2 = 3.69 ± 2.21; p ≤ 0.001; d = 0.37 ± 0.16 Y2 - Y3 = 1.28 ± 2.94; p = 0.76; d = 0.10 ± 0.16 Y3 - Y4 = −2.52 ± 3.56; p = 0.3; d = −0.16 ± 0.16
Running Back	87.51 ± 2.89 (33)	89.28 ± 3.12 (22)	88.71 ± 3.49 (13)	91.02 ± 4.13 (7)	Y1 - Y2 = 1.77 ± 2.93; p = 0.4; d = 0.12 ± 0.16 Y2 - Y3 = −0.57 ± 3.74; p = 0.98; d = −0.03 ± 0.16 Y3 - Y4 = 2.31 ± 5.07; p = 0.65; d = 0.10 ± 0.16
Safety	86.28 ± 2.44 (48)	88.8 ± 2.73 (27)	90.13 ± 2.99 (18)	88.69 ± 3.59 (9)	Y1 - Y2 = 2.52 ± 2.55; p = 0.05; d = 0.21 ± 0.16 Y2 - Y3 = 1.33 ± 3.16; p = 0.7; d = 0.09 ± 0.16 Y3 - Y4 = −1.44 ± 4.21; p = 0.81; d = −0.07 ± 0.16
Specialist	79.29 ± 3.19 (28)	82.47 ± 3.48 (18)	83.61 ± 3.78 (12)	89.3 ± 4.3 (7)	Y1 - Y2 = 3.18 ± 3.16; p = 0.05; d = 0.21 ± 0.16 Y2 - Y3 = 1.14 ± 3.76; p = 0.86; d = 0.06 ± 0.16 Y3 - Y4 = 5.69 ± 4.78; p = 0.01; d = 0.25 ± 0.16
Tight End	78.49 ± 3.14 (29)	80.78 ± 3.39 (19)	80.65 ± 3.74 (12)	81.09 ± 4.47 (6)	Y1 - Y2 = 2.3 ± 3.05; p = 0.21; d = 0.16 ± 0.16 Y2 - Y3 = −0.13 ± 3.86; p = 1; d = −0.01 ± 0.16 Y3 - Y4 = 0.44 ± 5.09; p = 1; d = 0.02 ± 0.16
All	80.96 ± 1.03 (478)	83.61 ± 1.07 (331)	83.98 ± 1.15 (193)	84.3 ± 1.28 (106)	Y1 - Y2 = 2.65 ± 0.81; p ≤ 0.001; d = 0.71 ± 0.16 Y2 - Y3 = 0.37 ± 1.01; p = 0.86; d = 0.08 ± 0.16 Y3 - Y4 = 0.32 ± 1.36; p = 0.97; d = 0.05 ± 0.16

Est: Estimated marginal means; 95% CL: 95% confidence limit; *n*: sample; *d*: Cohen's *d* effect size; Y1: year 1; Y2: year 2; Y3: year 3; Y4: year 4.

Table 5. Broad jump of NCAA Division I American football players over a 4-year collegiate career by position.

Position	Year 1 Est. ± 95% CL (n)	Year 2 Est. ± 95% CL (n)	Year 3 Est. ± 95% CL (n)	Year 4 Est. ± 95% CL (n)	Between year contrasts Est. ± 95% CL; *p* value; d ± 95% CL
Broad Jump (m)					
Corner	2.94 ± 0.05 (43)	2.95 ± 0.06 (30)	2.96 ± 0.07 (13)	2.89 ± 0.07 (10)	Y1 - Y2 = 0.01 ± 0.05; p = 0.97; d = 0.04 ± 0.16 Y2 - Y3 = 0.02 ± 0.07; p = 0.91; d = 0.06 ± 0.16 Y3 - Y4 = −0.07 ± 0.08; p = 0.1; d = −0.19 ± 0.17
Defensive Ends	2.78 ± 0.06 (36)	2.83 ± 0.06 (25)	2.85 ± 0.07 (14)	2.79 ± 0.08 (6)	Y1 - Y2 = 0.05 ± 0.05; p = 0.12; d = 0.18 ± 0.16 Y2 - Y3 = 0.02 ± 0.07; p = 0.84; d = 0.07 ± 0.16 Y3 - Y4 = −0.05 ± 0.1; p = 0.47; d = −0.12 ± 0.16
Defensive Tackles	2.56 ± 0.06 (38)	2.61 ± 0.06 (29)	2.61 ± 0.06 (17)	2.64 ± 0.08 (7)	Y1 - Y2 = 0.05 ± 0.05; p = 0.02; d = 0.24 ± 0.16 Y2 - Y3 = 0 ± 0.06; p = 1; d = −0.01 ± 0.17 Y3 - Y4 = 0.03 ± 0.09; p = 0.87; d = 0.06 ± 0.16
Linebacker	2.8 ± 0.04 (56)	2.84 ± 0.05 (35)	2.87 ± 0.05 (20)	2.82 ± 0.06 (12)	Y1 - Y2 = 0.05 ± 0.04; p = 0.03; d = 0.23 ± 0.16 Y2 - Y3 = 0.03 ± 0.05; p = 0.52; d = 0.11 ± 0.16 Y3 - Y4 = −0.05 ± 0.07; p = 0.27; d = −0.15 ± 0.16
Offensive Line	2.47 ± 0.04 (83)	2.51 ± 0.04 (59)	2.53 ± 0.04 (38)	2.53 ± 0.06 (12)	Y1 - Y2 = 0.04 ± 0.04; p = 0.03; d = 0.24 ± 0.16 Y2 - Y3 = 0.02 ± 0.04; p = 0.65; d = 0.11 ± 0.16 Y3 - Y4 = 0 ± 0.07; p = 1; d = −0.01 ± 0.16
Quarterback	2.77 ± 0.07 (23)	2.87 ± 0.08 (16)	2.9 ± 0.08 (9)	2.8 ± 0.1 (4)	Y1 - Y2 = 0.1 ± 0.06; p ≤ 0.001; d = 0.32 ± 0.16 Y2 - Y3 = 0.03 ± 0.08; p = 0.83; d = 0.07 ± 0.16 Y3 - Y4 = −0.1 ± 0.12; p = 0.15; d = −0.18 ± 0.16
Receiver	2.89 ± 0.04 (67)	2.96 ± 0.05 (44)	2.98 ± 0.05 (17)	2.96 ± 0.06 (16)	Y1 - Y2 = 0.07 ± 0.04; p ≤ 0.001; d = 0.37 ± 0.16 Y2 - Y3 = 0.02 ± 0.06; p = 0.73; d = 0.09 ± 0.16 Y3 - Y4 = −0.02 ± 0.07; p = 0.8; d = −0.08 ± 0.16
Running Back	2.82 ± 0.06 (34)	2.85 ± 0.06 (21)	2.87 ± 0.07 (13)	2.88 ± 0.08 (7)	Y1 - Y2 = 0.03 ± 0.06; p = 0.52; d = 0.11 ± 0.16 Y2 - Y3 = 0.02 ± 0.07; p = 0.87; d = 0.06 ± 0.16 Y3 - Y4 = 0 ± 0.09; p = 1; d = 0.01 ± 0.16
Safety	2.85 ± 0.05 (48)	2.91 ± 0.06 (26)	2.9 ± 0.06 (19)	2.84 ± 0.07 (8)	Y1 - Y2 = 0.06 ± 0.05; p = 0.02; d = 0.24 ± 0.16 Y2 - Y3 = −0.01 ± 0.06; p = 0.98; d = −0.03 ± 0.16 Y3 - Y4 = −0.06 ± 0.08; p = 0.25; d = −0.16 ± 0.16
Specialist	2.68 ± 0.06 (28)	2.73 ± 0.07 (17)	2.77 ± 0.08 (11)	2.82 ± 0.08 (7)	Y1 - Y2 = 0.05 ± 0.06; p = 0.15; d = 0.17 ± 0.16 Y2 - Y3 = 0.04 ± 0.07; p = 0.61; d = 0.10 ± 0.17 Y3 - Y4 = 0.06 ± 0.09; p = 0.38; d = 0.14 ± 0.17
Tight End	2.76 ± 0.06 (28)	2.75 ± 0.07 (18)	2.77 ± 0.07 (12)	2.78 ± 0.09 (6)	Y1 - Y2 = −0.01 ± 0.06; p = 0.99; d = −0.02 ± 0.16 Y2 - Y3 = 0.02 ± 0.07; p = 0.94; d = 0.05 ± 0.16 Y3 - Y4 = 0.01 ± 0.1; p = 0.99; d = 0.02 ± 0.17
All	2.74 ± 0.02 (488)	2.79 ± 0.02 (325)	2.8 ± 0.02 (184)	2.78 ± 0.03 (95)	Y1 - Y2 = 0.04 ± 0.02; p ≤ 0.001; d = 0.61 ± 0.16 Y2 - Y3 = 0.02 ± 0.02; p = 0.2; d = 0.18 ± 0.16 Y3 - Y4 = −0.02 ± 0.03; p = 0.2; d = −0.18 ± 0.16

Est: Estimated marginal means; 95% CL: 95% confidence limit; *n*: sample; *d*: Cohen's *d* effect size; Y1: year 1; Y2: year 2; Y3: year 3; Y4: year 4.

Table 6. 40-yard dash of NCAA Division I American football players over a 4-year collegiate career by position.

Position	Year 1 Est. ± 95% CL (n)	Year 2 Est. ± 95% CL (n)	Year 3 Est. ± 95% CL (n)	Year 4 Est. ± 95% CL (n)	Between year contrasts Est. ± 95% CL; *p* value; d ± 95% CL
40-yard sprint (s)					
Corner	4.54 ± 0.06 (43)	4.53 ± 0.06 (31)	4.56 ± 0.07 (13)	4.53 ± 0.08 (10)	Y1 - Y2 = −0.02 ± 0.02; p = 0.78; d = −0.08 ± 0.16 Y2 - Y3 = 0.03 ± 0.03; p = 0.7; d = 0.09 ± 0.16 Y3 - Y4 = −0.02 ± 0.03; p = 0.89; d = −0.06 ± 0.16
Defensive Ends	4.83 ± 0.06 (38)	4.81 ± 0.07 (14)	4.81 ± 0.09 (6)	4.65 ± 0.18 (4)	Y1 - Y2 = −0.04 ± 0.02; p = 0.3; d = −0.14 ± 0.15 Y2 - Y3 = 0.02 ± 0.03; p = 0.92; d = 0.05 ± 0.16 Y3 - Y4 = 0 ± 0.04; p = 1; d = −0.01 ± 0.16
Defensive Tackles	5.1 ± 0.06 (37)	5.06 ± 0.06 (31)	5.07 ± 0.07 (18)	5.08 ± 0.08 (9)	Y1 - Y2 = −0.03 ± 0.02; p = 0.35; d = −0.13 ± 0.16 Y2 - Y3 = 0.01 ± 0.02; p = 0.99; d = 0.03 ± 0.16 Y3 - Y4 = 0.01 ± 0.03; p = 1; d = 0.01 ± 0.16
Linebacker	4.76 ± 0.05 (60)	4.73 ± 0.05 (37)	4.7 ± 0.06 (23)	4.73 ± 0.07 (13)	Y1 - Y2 = −0.04 ± 0.02; p = 0.15; d = −0.17 ± 0.16 Y2 - Y3 = −0.03 ± 0.02; p = 0.57; d = −0.10 ± 0.16 Y3 - Y4 = 0.04 ± 0.03; p = 0.6; d = 0.10 ± 0.16
Offensive Line	5.34 ± 0.04 (85)	5.29 ± 0.04 (62)	5.28 ± 0.05 (41)	5.25 ± 0.06 (15)	Y1 - Y2 = −0.04 ± 0.01; p = 0.01; d = −0.25 ± 0.16 Y2 - Y3 = −0.01 ± 0.02; p = 0.88; d = −0.06 ± 0.16 Y3 - Y4 = −0.03 ± 0.03; p = 0.69; d = −0.09 ± 0.16
Quarterback	4.76 ± 0.08 (23)	4.69 ± 0.08 (18)	4.7 ± 0.09 (9)	4.75 ± 0.11 (4)	Y1 - Y2 = −0.07 ± 0.03; p = 0.05; d = −0.21 ± 0.16 Y2 - Y3 = 0 ± 0.03; p = 1; d = 0.00 ± 0.16 Y3 - Y4 = 0.05 ± 0.05; p = 0.73; d = 0.08 ± 0.16
Receiver	4.62 ± 0.05 (67)	4.57 ± 0.05 (44)	4.56 ± 0.06 (23)	4.56 ± 0.06 (17)	Y1 - Y2 = −0.05 ± 0.02; p = 0.03; d = −0.22 ± 0.16 Y2 - Y3 = −0.02 ± 0.02; p = 0.83; d = −0.07 ± 0.16 Y3 - Y4 = 0 ± 0.03; p = 1; d = 0.01 ± 0.16
Running Back	4.66 ± 0.06 (34)	4.62 ± 0.07 (23)	4.6 ± 0.07 (16)	4.62 ± 0.09 (7)	Y1 - Y2 = −0.03 ± 0.02; p = 0.46; d = −0.12 ± 0.15 Y2 - Y3 = −0.02 ± 0.03; p = 0.88; d = −0.06 ± 0.16 Y3 - Y4 = 0.02 ± 0.04; p = 0.97; d = 0.03 ± 0.16
Safety	4.64 ± 0.06 (48)	4.87 ± 0.08 (13)	4.83 ± 0.1 (6)	2.84 ± 0.07 (8)	Y1 - Y2 = −0.02 ± 0.02; p = 0.78; d = −0.07 ± 0.15 Y2 - Y3 = −0.02 ± 0.02; p = 0.85; d = −0.06 ± 0.16 Y3 - Y4 = 0.04 ± 0.03; p = 0.7; d = 0.09 ± 0.16
Specialist	4.82 ± 0.07 (28)	4.8 ± 0.08 (19)	4.8 ± 0.08 (12)	4.78 ± 0.09 (7)	Y1 - Y2 = −0.02 ± 0.02; p = 0.82; d = −0.07 ± 0.16 Y2 - Y3 = 0 ± 0.03; p = 1; d = −0.01 ± 0.16 Y3 - Y4 = −0.02 ± 0.04; p = 0.95; d = −0.04 ± 0.16
Tight End	4.86 ± 0.07 (29)	4.85 ± 0.08 (19)	4.87 ± 0.08 (13)	4.83 ± 0.1 (6)	Y1 - Y2 = −0.02 ± 0.02; p = 0.91; d = −0.05 ± 0.16 Y2 - Y3 = 0.03 ± 0.03; p = 0.78; d = 0.08 ± 0.16 Y3 - Y4 = −0.05 ± 0.04; p = 0.63; d = −0.10 ± 0.16
All	4.84 ± 0.03 (496)	4.81 ± 0.03 (344)	4.81 ± 0.03 (203)	4.81 ± 0.03 (103)	Y1 - Y2 = −0.03 ± 0.02; p ≤ 0.001; d = −0.44 ± 0.16 Y2 - Y3 = 0 ± 0.02; p = 0.97; d = −0.03 ± 0.15 Y3 - Y4 = 0 ± 0.03; p = 1; d = 0.00 ± 0.15

Est: Estimated marginal means; 95% CL: 95% confidence limit; *n*: sample; *d*: Cohen's *d* effect size; Y1: year 1; Y2: year 2; Y3: year 3; Y4: year 4.

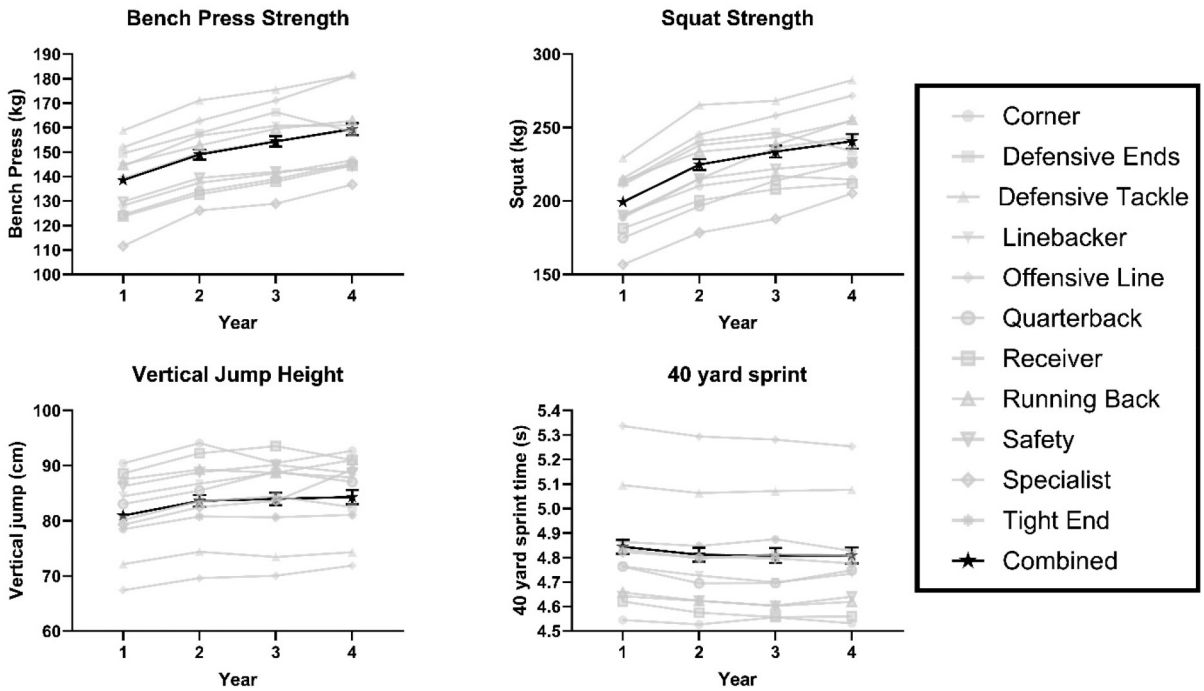

Figure 1. Estimated marginal means for the bench press, squat, vertical jump, and 40-yard sprint by positional group and combined, by year.

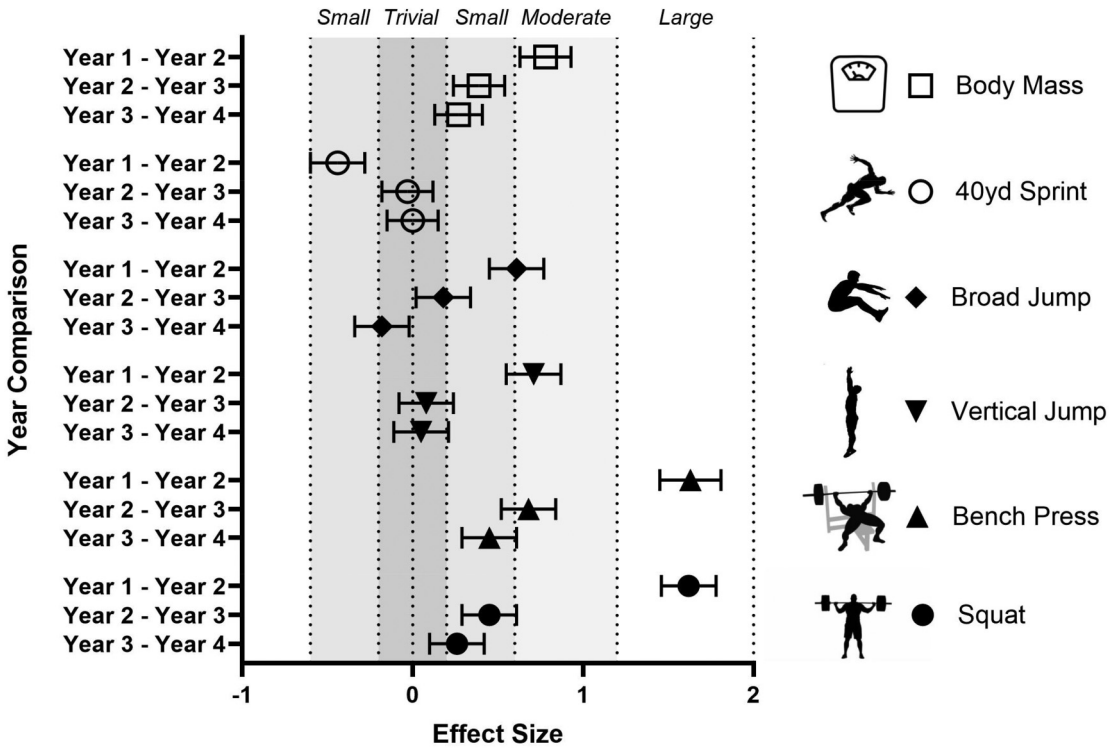

Figure 2. Annual changes in physical qualities presented as effect size ± 95% confidence intervals.

and conditioning programmes), but this study is the first to demonstrate this occurrence as youth players transition into the collegiate level and experience near professional levels of training and support in American football.

While this study is the first to detail the longitudinal changes in strength, power, speed, and body mass of collegiate Division I American football athletes by positional group, there are limitations. First, while standardised training programmes were provided (which involved consistent themes and regularity) and led by the same strength and conditioning coaches, small changes were made year to year. These changes are a consequence of measuring real-world athletes in-situ and likely had trivial effects on the overall variability in rates of adaptations across years. Second, it is feasible that a form of 'reverse survivorship bias' could have occurred which may have reduced the estimated mean of physical qualities in the latter years. It is plausible that athletes with the greatest physical qualities declared for the National Football League draft across this 15-year period, with numerous players from this collegiate programme being selected. This may have caused a very minor reduction in the overall reported qualities in the latter years (e.g. years 3 and 4) as the best players opted for professional status.

In conclusion, this study demonstrates the longitudinal changes in strength, speed, power, and body mass characteristics of collegiate Division 1 American football players across different positional groups. It is clear that as athletes age and are exposed to greater amounts of high-quality training, there are notable improvements in physical qualities. However, the rate of change across years substantially reduces as improvements become harder to attain. This is likely due to the novel intensive stimulus that is provided to athletes as they first enter the collegiate system and the diminishing returns that often occur as athletes adapt. This is of importance for coaches and athletes, as it can help guide expectations and justify why improvements in performance may be appearing to slow down. To help mitigate these reductions in adaptation across years, greater personalisation of training prescription could be recommended alongside increasing the emphasis on transferring these physical qualities to on-field performance.

Practical applications

American football requires athletes to have well-developed physical qualities, with positional groups requiring greater development of certain traits (e.g. offensive linemen require high levels of upper and lower body strength). The findings from this study provide normative data that can be used to support training targets and recommendations. For example, freshmen offensive linemen wishing to compete in Division 1 NCAA football should be aiming to back squat >200 kg and have a body mass of >125 kg. Alternatively, a wide receiver may not prioritise their body mass but may wish to focus on improving their speed. Additionally, the findings from this paper demonstrate the expected annual improvements that occur in physical qualities, with smaller improvements occurring year-on-year. This is important for athletes and coaches as it aids in the development of realistic goals and can be used to explain why athletes appear to plateau as training time accumulates. While speculative, it might be beneficial for more advanced players to have greater individualisation in their training programmes so that continual improvement can occur. Additionally, the findings suggest a need for young athletes (e.g. high school football players) to continue investing time into strength and conditioning so that they are competitive with more physically developed older athletes.

Consent for publication

Written informed consent for publication was obtained from all athletes.

Consent to participate

Written informed consent was obtained from all athletes.

Data availability

The datasets used and/or analysed during the current study can be made available by the corresponding author on reasonable request.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Ethical considerations

The protocol was approved by the University of Missouri Institutional Review Board in accordance with the Declaration of Helsinki.

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